

Hadoop HDFS and MapReduce

Hadoop is an open-source framework designed to store and process massive datasets across distributed computing systems. It is highly scalable and fault-tolerant, making it an essential tool for big data processing in industries like agriculture, finance, and logistics.

HDFS (Hadoop Distributed File System)

HDFS is the storage layer of Hadoop and is designed to handle large datasets efficiently.

Key Features of HDFS:

1. Scalability: HDFS can scale horizontally by adding more nodes to the system.
2. Fault Tolerance: Data is stored in multiple copies (replication factor) to ensure reliability.
3. High Throughput: Optimized for batch processing of large files rather than lowlatency operations.
4. Master-Slave Architecture:
 - **NameNode:** Manages metadata and the directory structure of the file system.
 - **DataNodes:** Store the actual data in blocks and handle read/write operations.

Workflow in HDFS:

- Data is split into blocks (default size is 128 MB or 64 MB).
- Blocks are distributed across multiple DataNodes in the cluster.
- A replication mechanism ensures redundancy.

MapReduce Framework

MapReduce is the processing engine of Hadoop, designed for parallel computation on large datasets. It follows the divide-and-conquer principle.

Key Components:

1. Map Phase:

- Splits data into key-value pairs.
- Processes each pair independently.
- Example: Counting words in a document (Word, Count).

2. Shuffle and Sort Phase:

- Organizes and groups intermediate key-value pairs for further processing.

3. Reduce Phase:

- Aggregates the grouped data into meaningful results.
- Example: Summing up the word counts.

Example 1. *Counting the frequency of words in a text*

1. *Input: A large document split into smaller chunks.*
2. *Map Output: ("apple", 1), ("banana", 1), ("apple", 1).*
3. *Reduce Output: ("apple", 2), ("banana", 1).*

ByteMonk (2023). Map Reduce explained with example — System Design.

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/cHGaQz0E7AU>



uFaber (2015). Hadoop MapReduce Examples - Hadoop MapReduce tutorials - Hadoop

Video Visit the URL below to view a video:

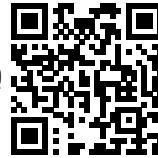
<https://www.youtube.com/embed/akjVm-yDPN8>



internet-class (2016). What is MapReduce?.

Video Visit the URL below to view a video:

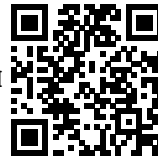
<https://www.youtube.com/embed/43fqzaSH0CQ>



0.1 Udacity (2015). HDFS - Intro to Hadoop and MapReduce.

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/vdkx2xasGIM>



Simplilearn (20210). Hadoop In 5 Minutes — What Is Hadoop? — Introduction To Hadoop — Hadoop Explained

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/aReuLtY0YMI>



IvyProSchool (2023). Hadoop HDFS Commands and MapReduce with Example: Step-byStep Guide — Hadoop Tutorial — IvyProSchool.

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/7056u3LyPTY>

